Python Programming Interview Notes

Python fundamentals, collections, functions, OOP, and exceptions

Python Fundamentals

- Python is a high-level, general-purpose language. Indentation defines code blocks.
- Common built-in types include int, float, bool, str, list, tuple, set, and dict.
- Python is dynamically typed: a variable can refer to objects of different types during its lifetime.
- Interview prompts: What is Python? What are mutable and immutable objects?

Collections

- A list is ordered and mutable. A tuple is ordered and immutable. A set stores unique elements. A dictionary maps keys to values.
- Lists use square brackets, tuples commonly use parentheses, sets use braces with elements, and dictionaries use key:value pairs.
- Interview prompts: Compare a list and tuple. When would you use a set? What makes a valid dictionary key?

Functions and Scope

- Functions are defined with def and can accept parameters and return values.
- Python supports positional and keyword arguments, default parameters, *args, and **kwargs.
- Local names are defined in a function scope; global and nonlocal affect name binding in other scopes.
- Interview prompts: Parameters versus arguments? What are lambda functions?

Object-Oriented Programming

- A class defines a type and an object is an instance. __init__ commonly initializes instance attributes.
- Inheritance allows a class to extend or specialize another class. Polymorphism lets different objects support a common operation.
- Instance methods receive self by convention. Class methods use @classmethod and receive cls; static methods use @staticmethod.
- Interview prompts: What is self? Explain inheritance. Compare instance, class, and static methods.

Exceptions and Files

- Use try and except to handle exceptions. else runs if try completes without an exception; finally is commonly used for cleanup.
- The with statement manages resources such as files, ensuring they are closed appropriately.
- Use open(path, mode, encoding='utf-8') for text files. Common modes include r, w, and a.
- Interview prompts: How does exception handling work? Why use with when opening a file?

Iterators and Coding Practice

- An iterator provides values one at a time. A generator function uses yield to produce values lazily.
- A list comprehension creates a list from an expression and iterable, optionally applying a condition.
- Practice: palindrome checks, Fibonacci sequences, prime numbers, frequency counts, sorting dictionaries, list deduplication, and word counts.
- Explain each solution with examples, edge cases, and time complexity.